# Chemistry Formula Sheet (College & Pre-Med)

General and organic chemistry for college, pre-med and MCAT review, plus the clinical chemistry med school builds on

MindShelf

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## Units & Concentration

$$M = \frac{n}{V_{soln}} \quad m = \frac{n_{solute}}{m_{solvent\ (kg)}} \quad n = \frac{m}{\mathcal{M}}$$

$C_1V_1 = C_2V_2$ eq = $n \times |z|$ osmolarity = $i \times M$

$\mathcal{M}$ molar mass; $i$ van 't Hoff factor; $z$ ion charge. *E.g.* 500 mL of 0.1 M from 2 M stock: $V_1 = 0.1 \times 500/2 = 25$ mL.

## Atoms & Periodic Trends

$$A = Z + N \quad E = h\nu = \frac{hc}{\lambda} \quad c = \lambda\nu$$

$n \geq 1$; $\ell = 0 \ldots n-1$; $m_\ell = -\ell \ldots +\ell$; $m_s = \pm\frac{1}{2}$.

Across a period →: radius ↓, ionization energy ↑, electronegativity ↑; down a group ↓: reversed.

## Bonding & Geometry

### Formal charge

$$FC = V - N - \frac{B}{2}$$

$V$ valence, $N$ nonbonding, $B$ bonding e⁻. Best: charges near zero, minus on the more electronegative atom.

| Dom. | LP | Shape (e.g.) | Angle | Hybrid |
|---|---|---|---|---|
| 2 | 0 | linear ($CO_2$) | 180° | sp |
| 3 | 0 | trig. planar ($BF_3$) | 120° | $sp^2$ |
| 3 | 1 | bent ($SO_2$) | <120° | $sp^2$ |
| 4 | 0 | tetrahedral ($CH_4$) | 109.5° | $sp^3$ |
| 4 | 1 | trig. pyramidal ($NH_3$) | ≈107° | $sp^3$ |
| 4 | 2 | bent ($H_2O$) | ≈104.5° | $sp^3$ |
| 5 | 0 | trig. bipyramid ($PCl_5$) | 90, 120° | $sp^3d$ |
| 6 | 0 | octahedral ($SF_6$) | 90° | $sp^3d^2$ |

**IMFs:** ion–dipole > H-bond (H on N, O, F) > dipole–dipole > London. Stronger IMFs: higher bp, lower vapor pressure.

## Gases

$$PV = nRT \quad \frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$P_i = X_iP_{tot} \quad \frac{r_1}{r_2} = \sqrt{\frac{\mathcal{M}_2}{\mathcal{M}_1}}$$

$T$ in kelvin. *E.g.* 2 atm, 5 L, 300 K, $R$ = 0.08206: $n = PV/RT = 0.406$ mol.

## Thermochemistry

$$q = mc\Delta T \quad \Delta G = \Delta H - T\Delta S$$

$$\Delta H_{rxn} = \sum \Delta H_f^{prod} - \sum \Delta H_f^{react}$$

$$\Delta G^\circ = -RT \ln K = -nFE^\circ$$

*E.g.* $\Delta H = -92.2$ kJ/mol, $\Delta S = -0.199$ kJ/(mol·K), 298 K: $\Delta G = -32.9$ kJ/mol.

Spontaneous ($\Delta G < 0$): $\Delta H < 0, \Delta S > 0$ always; $\Delta H > 0, \Delta S < 0$ never; both negative at low $T$; both positive at high $T$.

## Kinetics & Enzymes

$$\text{rate} = k[A]^m[B]^n \quad k = Ae^{-E_a/RT}$$

First order: $\ln[A] = \ln[A]_0 - kt$; $t_{1/2} = 0.693/k$.

$$v = \frac{V_{max}[S]}{K_m + [S]} \quad \frac{1}{v} = \frac{K_m}{V_{max}} \cdot \frac{1}{[S]} + \frac{1}{V_{max}}$$


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Orders come from experiment, not coefficients. *E.g.* $K_m$ = 2 mM, [S] = 6 mM, $V_{max}$ = 100: $v$ = 75 µmol/min.

| Inhibitor | Binds | $K_m$ | $V_{max}$ |
|---|---|---|---|
| Competitive | E (active site) | ↑ | = |
| Noncompetitive | E and ES equally | = | ↓ |
| Uncompetitive | ES only | ↓ | ↓ |
| Mixed | E or ES, unequally | ↑ / ↓ | ↓ |

# Equilibrium & Solubility

$$K = \frac{[C]^c[D]^d}{[A]^a[B]^b} \qquad K_{sp}\left(M_xX_y\right) = [M]^x[X]^y$$

$Q < K$: shifts right; $Q > K$: shifts left; solid forms if $Q > K_{sp}$. Common ion: lower solubility. Solids and pure liquids stay out of $K$.

# Acids, Bases & Buffers

$$\text{pH} = -\log[H_3O^+] \qquad K_w = [H_3O^+][OH^-]$$

$$pK_a + pK_b = pK_w \qquad \text{pH} = pK_a + \log\frac{[A^-]}{[HA]}$$

Best buffer within $pK_a \pm 1$. Weak acid titrated by strong base: half-equivalence pH = $pK_a$; equivalence pH > 7.

| Disorder | pH | Primary | Comp. |
|---|---|---|---|
| Metabolic acidosis | ↓ | ↓ $HCO_3^-$ | ↓ $PaCO_2$ |
| Metabolic alkalosis | ↑ | ↑ $HCO_3^-$ | ↑ $PaCO_2$ |
| Respiratory acidosis | ↓ | ↑ $PaCO_2$ | ↑ $HCO_3^-$ |
| Respiratory alkalosis | ↑ | ↓ $PaCO_2$ | ↓ $HCO_3^-$ |

**CLINICAL NOTE Bicarbonate buffer**

$$\text{pH} = 6.1 + \log\frac{[HCO_3^-]}{0.03 \times P_aCO_2}$$

$HCO_3^-$ 24 mmol/L, $PaCO_2$ 40 mmHg: ratio 20, log 20 = 1.30103, pH 7.40.

# Solutions & Colligatives

$$\Delta T_f = iK_fm \qquad \Delta T_b = iK_bm \qquad \Pi = iMRT$$

Raoult: $P_{soln} = X_{solvent}P°_{solvent}$. Water: $K_f$ = 1.86 °C·kg/mol, $K_b$ = 0.512 °C·kg/mol. *E.g.* 0.5 mol/kg NaCl: $\Delta T_f = 2 \times 1.86 \times 0.5 = 1.86$ °C.

# Electrochemistry

$$E°_{cell} = E°_{cathode} - E°_{anode} \qquad \Delta G° = -nFE°$$

$$E = E° - \frac{RT}{nF}\ln Q = E° - \frac{0.0592}{n}\log Q$$

(25 °C, V). *E.g.* $E°$ = 1.1 V, $n$ = 2, $Q$ = 10: $E$ = 1.07 V.

Oxidation at the anode. **Galvanic:** $\Delta G < 0$, $E_{cell} > 0$, anode (−). **Electrolytic:** all three reversed.

**CLINICAL NOTE Membrane potential (37 °C)**

$$E_{ion} = \frac{61.5}{z}\log\frac{[X]_{out}}{[X]_{in}}\ \text{(mV)}$$

$K^+$ 4 mM out, 140 mM in: $E = 61.5 \times (-1.54407) = -95$ mV.

# Organic Essentials

**R/S:** rank by atomic number at the first difference (CIP), point the lowest priority away, trace 1 → 2 → 3: clockwise *R*, counterclockwise *S* (reverse if it points toward you).

| | $S_N1$ | $S_N2$ |
|---|---|---|
| Substrate | 3° > 2° | methyl > 1° > 2° |
| Rate law | $k$[RX] | $k$[RX][$Nu^-$] |
| Nucleophile | weak is fine | strong |
| Solvent | polar protic | polar aprotic |
| Stereo | racemization | inversion |

**Amino acid pI** (neutral side chain): pI = $\left(pK_{a1} + pK_{a2}\right)/2$; net charge + below pI, − above. *E.g.* 2.34 and 9.6: pI = 5.97.

# Spectroscopy

**Beer–Lambert**

$$A = -\log T = \varepsilon lc$$

*E.g.* $A$ = 0.5, $\varepsilon$ = 5,000 L/(mol·cm), $l$ = 1 cm: $c$ = 100 µM.

**CLINICAL NOTE Pulse oximetry**

$HbO_2$ and Hb absorb 660 nm and 940 nm light differently; the absorbance ratio gives $SpO_2$.

| IR band | $cm^{-1}$ |
|---|---|
| O–H (alcohol), broad | 3,200–3,600 |
| O–H (carboxylic acid), very broad | 2,500–3,100 |
| N–H, 1 or 2 peaks | 3,300–3,500 |
| C≡N / C≡C | 2,100–2,260 |
| C=O, strong & sharp | 1,670–1,780 |

| $^1$H NMR environment | δ (ppm) |
|---|---|
| alkyl C–H | 0.9–1.8 |

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| $^1$H NMR environment | δ (ppm) |
|---|---|
| H–C–C=O | 2.0–2.5 |
| H–C–O / H–C–X | 3.3–4.5 |
| vinylic =C–H | 4.5–6.5 |
| aromatic Ar–H | 6.5–8.0 |
| aldehyde CHO | 9.0–10.0 |
| carboxylic acid COOH, broad | 10–12 |

$^1$H NMR: $n$ neighboring H give $n + 1$ peaks; the integral is proportional to the number of H.

# High-Yield Med Chem

Anion gap = $Na^+ - (Cl^- + HCO_3^-)$
Osm = $2\ Na^+ + Glu/18 + BUN/2.8$
$CaO_2 = 1.34 \times Hb \times SaO_2 + 0.003 \times PaO_2$
$t_{1/2} = 0.693 \times V_d / CL$

Glu, BUN in mg/dL; Hb in g/dL. Reference: AG 4–12 mEq/L, osm 275–295 mOsm/kg intervals vary by lab and method.

**EXAMPLE**

- Na 140, Cl 100, $HCO_3^-$ 14: AG = 26 mEq/L (high).
- Na 140, glucose 90, BUN 14: Osm = 290 mOsm/kg.
- Hb 15, $SaO_2$ 0.98, $PaO_2$ 100: $CaO_2$ = 20.0 mL/dL, only 0.3 dissolved.
- $V_d$ 50 L, CL 5 L/h: $t_{1/2}$ = 6.9 h.

**CLINICAL NOTE Alveolar gas equation**

$$P_AO_2 = F_IO_2\left(P_{atm} - P_{H_2O}\right) - P_aCO_2 / \text{RQ}$$

Sea level 760 mmHg $P_{H_2O}$ 47 mmHg at 37 °C RQ ≈ 0.8.

# Constants & Traps

$N_A = 6.022 \times 10^{23}$ mol$^{-1}$ · $R = 8.314$ J/(mol·K) = 0.08206 L·atm/(mol·K) · $F = 96{,}485$ C/mol · $h = 6.626 \times 10^{-34}$ J·s · $c = 3.00 \times 10^8$ m/s · $K_w$ (25 °C) $= 1.0 \times 10^{-14}$ · 1 atm = 760 mmHg = 101.325 kPa · ideal gas at 0 °C, 1 atm: 22.4 L/mol · 0 °C = 273.15 K · water $c = 4.184$ J/(g·°C)

**TRAP**

- Use kelvin; match J and kJ in $\Delta H$, $\Delta S$.
- $M$ per L of solution; $m$ per kg of solvent.
- Colligative effects count particles ($i$).
- Catalysts lower $E_a$; only $T$ changes $K$.
- $\Delta G° = -RT \ln K$: with log, multiply by 2.303.
- At equilibrium $\Delta G = 0$ and $E_{cell} = 0$.

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